

William Bowley

Computational & Electrical Engineering Student @ RMIT University
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Summary

Computational and Electrical Engineering student specializing in electromagnetic systems, multi-physics simulation, and numerical tool development. Experienced developing open-source engineering software, simulation frameworks, and hardware prototypes for motor applications.

Skills

- **Languages:** Python, C/C++, JS, MATLAB
- **Simulation & Tools:** FEMM, LTspice, Fusion 360, Inventor, Altium, KiCad, PSIM (Basics)
- **Hardware:** STM32, PCB Design, Motor Design/Control, RS-485/CAN, Sensor Integration
- **Other:** Git, Linux, LaTeX/Markdown, FastAPI/Flask, React (basics)

Projects

OpenLSM

Aug 2025 – Present

Electrical Lead & Developer — Open Source Linear Synchronous Motor Project

- Developing transient multi-physics simulation model (electromagnetic, thermal, mechanical) for linear motor design and optimization.
- Developed a parametric motor geometry pipeline with NSGA-III multi-objective optimization capable of evaluating over 1,000 candidate designs.
- Designed, built, and validated a custom planar linear motor prototype with a three-phase bridge driver, demonstrating closed-loop linear motion.

PyFEA

Dec 2024 – Present

Project Lead — Multi-Physics Solver Framework

- Developing a multi-physics solver framework with an intermediate representation linking finite element and lumped parameter models through modular solver adaptors.

PicoUnits

Jan 2026 – Present

Project Lead — Dimensional Constraint System for Python

- Created PyPI package for runtime dimensional analysis and unit-aware engineering computations.
- Developed a domain-specific language enabling unit-aware engineering configuration files and runtime dimensional validation.

Experience

Powertrain Engineer

Apr 2026 – Present

RMIT Motorsport — Formula SAE Team (dyno-boards)

- Simulating Permanent Magnet Synchronous Motors (PMSMs) in PSIM for drivetrain analysis.
- Designed isolated dyno communication electronics and supported high-voltage system integration.
- Developed analysis framework for active vs passive precharge systems.

Hercules Challenge — IsoPod Project

Mar 2026 – Jul 2026

RMIT University (Isopod)

- Co-designed assistive axial shake generator for medication reminder system (0.45W @ 8.8Hz).
- Utilized custom FEA models for electromagnetic design with accuracy to within 12–22% of measured results.
- Worked in multidisciplinary team. Project reached finals and received Commendation Award.

Education

Bachelor of Engineering (Honours) — Electrical

Mar 2026 – Nov 2029

RMIT University, Melbourne

Victorian Certificate of Education (VCE)

2025

Auburn High School

Additional

- Competitive swimmer — 4 × state qualifier and 7 × regional qualifier in 50 m, 100 m, and 200 m Backstroke, and 200m Freestyle Relay
- Community volunteering — Supported primary school fundraising events through Bunnings sausage sizzles and constructed approximately 12 garden beds for Camberwell South Primary School.